$6 - 3 = 3$
$6 + 2 = 8$

Linear transformation Given by a matrix

Let A be an $m \times n$ matrix. the function T defind by

$$T(V) = AV$$

is a Linear transformation from R^n into R^m

$$A = \begin{bmatrix} 1 & 0 & -1 & 2 \\ 3 & 1 & 0 & 0 \end{bmatrix}_{2\times 4}$$

$T(V) = AV$

$\therefore$ T is a Linear transformation $R^4 \longrightarrow R^2$

If $T: R^2 \longrightarrow R^2$ by $T\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) = \begin{bmatrix} 2a - b \\ a + 3b \end{bmatrix}$

then $T\left(\begin{bmatrix} 2 \\ 2 \end{bmatrix}\right) = ??$

$$\begin{bmatrix} 2\times 2 - 2 \\ 2 + 3\times 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 8 \end{bmatrix}$$

singular ⟶ inverse ملهاش

المحدد = صفر

eigenvalue = صفر

invertable = non singular

inverse ليها

المحدد ≠ صفر

—

① 1

②

Part 2

REF , RREF

systems of Linear equation

① اي صف فيه اصفار لازم يكون في نهاية المصفوفة
حتى لو كان صفين يكون في نهاية المصفوفة

② المصفوفة تحتوي علي اصفار و واحد فقط

③ مينفعش تلاقي واحد الواحد تحت بعض لازم
يكونوا كده

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

مش لازم بالدور بس يكون كده

RREF

حاجة زيادة

① ان لو في 1 في الصف يكون تحته اصفار

مثال

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Part 8

Matrices

$\boxed{\text{Exp}}$

$$\begin{bmatrix} 2 & \cancel{8}\,5 & 3 & 7 \\ 2 & 3 & 1 & 0 \\ 6 & 1 & 9 & 4 \\ 6 & 4 & 0 & 7 \end{bmatrix}_{4\times 4}$$

$a_{13} = 3$

$a_{34} = 4$

① square matrix : rows = colmns

$$\begin{bmatrix} 3 & -1 & -3 \\ 2 & 4 & 0 \\ -1 & 5 & 6 \end{bmatrix}$$

main diagonal

② upper triangular matrix

$$\begin{bmatrix} 3 & -1 & -5 \\ 0 & 4 & 7 \\ 0 & 0 & 6 \end{bmatrix}$$

③ Lower triangular matrix

$$\begin{bmatrix} 3 & 0 & 0 \\ -2 & 4 & 0 \\ 2 & 7 & -6 \end{bmatrix}$$

④ Diagonal matrix

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & -6 \end{bmatrix}$$

⑤ Identity matrix

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

⑥ zero matrix

$$O_{2\times 2} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

⑦ column matrix (n - vector)

$\begin{bmatrix} 2 \\ -4 \\ 6 \end{bmatrix}_{3\times 1}$ is a 3 - column vector

⑧ Row matrix (n- vector

$\underset{1\times 3}{\begin{bmatrix} 3 & 5 & 8 \end{bmatrix}}$ is a - row vector

⑨ orthogonal

$$A^{-1} = A^{T}$$

$$A^{-1}A = AA^{T} = I$$

$(A^T)^T = A$

$(A+B)^T = A^T + B^T$

$(CA)^T = C(A^T)$

C constant

$(AB)^T = B^T A^T$

$(I)^T = I$

symmetric $\quad A^T = A$

$$C = \begin{bmatrix} 1 & -1 & 3 & 4 \\ -1 & 8 & 5 & 6 \\ 3 & 5 & 9 & -2 \\ 4 & 6 & -2 & 0 \end{bmatrix}$$

$$C^T = \begin{bmatrix} 1 & -1 & 3 & 4 \\ -1 & 8 & 5 & 6 \\ 3 & 5 & 9 & -2 \\ 4 & 6 & -2 & 0 \end{bmatrix}$$

$C^T = C$

symmetric matrices

skew symmetric matrices

$$A^T = -A$$

$$C = \begin{bmatrix} 0 & -1 & -3 & 4 \\ 1 & 0 & 5 & -6 \\ 3 & -5 & 0 & -2 \\ -4 & 6 & 2 & 0 \end{bmatrix}$$

$$C^T = \begin{bmatrix} 0 & 1 & 3 & -4 \\ -1 & 0 & -5 & 6 \\ -3 & 5 & 0 & 2 \\ 4 & -6 & -2 & 0 \end{bmatrix}$$

Part 11

An $n \times n$ matrix A is invertible or (nonsingular)

$AB = BA = I$

The Inverse of A Matrix

~~Anf~~ A matrix that doesn't have an invers is noninverible (or singular)

The invers of A is denoted by A^{-1}

$$AA^{-1} = A^{-1}A = I$$

اجيب ال inverse ازاي

① نضيف ليها مصفوفة الوحدة I

② اجيب rref

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ -6 & 2 & 3 \end{bmatrix}$$

Find the inverse of the matrix

$$\begin{array}{c} \quad A \qquad\qquad\qquad I \\ \left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 1 & 0 & 0 \\ 1 & 0 & -1 & 0 & 1 & 0 \\ -6 & 2 & 3 & 0 & 0 & 1 \end{array}\right] \end{array} \xrightarrow[R_3 = 6R_1 + R_3]{R_2 = -R_1 + R_2}$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & \frac{-2}{7} & \frac{3}{7} & \frac{-1}{7} & 0 \\ 0 & 7 & -2 & 2 & 0 & 1 \end{array}\right] \xrightarrow{R_3 = -7R_2 + R_3}$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & \frac{-2}{7} & \frac{-3}{7} & \frac{-1}{7} & 0 \\ \boxed{0 \quad 0 \quad 0} & & & \underbrace{-1 \quad 1 \quad 1}_{\neq A^{-1}} & & \end{array}\right]$$

$\neq I$

A has no Inverse or is nonivertible
is sigular

If A is a 2X2 matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

A is invertible if and only if $ad - bc \neq 0$

$$A = \begin{bmatrix} 3 & -1 \\ -2 & 2 \end{bmatrix}$$

$$ad - bc = (3)(2) - (-1)(-2) = 4$$

$$A^{-1} = \frac{1}{4}\begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & \frac{3}{4} \end{bmatrix}$$

Note:

The inverse of a matrix in unique

If A and B are non singular then $(A^{-1})^{-1} = A$

$$(A^T)^{-1} = (A^{-1})^T$$

$$(AB)^{-1} = B^{-1}A^{-1}$$

$$(A^n)^{-1} = \frac{1}{n} A^{-1}$$

$$(I)^{-1} = I$$

$$(A+B)^{-1} \neq A^{-1} + B^{-1}$$

$$(A+B)^{-1} = A^{-1} - A^{-}BA^{-1} + A^{+1}BAA^{-1}BA^{-1}$$

/ /

(Part 13)

Periodic matrix

$A^k = I$ For $k \in \mathbb{Z}^+$ (positive) integers

The smallest such k is the order

اى اقل قيمة ممكنة تديلها لما تضرب المصفوفة كلها = يوصل لمصفوفة الوحده

Ex

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \longrightarrow A^4 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

A is periodical with order $= 4$

prove that matrix is periodical

$$A = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

الضرب الصف x عمود

$$A^2 = AA = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

$$A^3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I_{3\times3}$$

A is Periodic with order $= 3$

/ /

Part 16

Determinants

upper Triangular
Lower Triangular
diagonal matrix

$\det(A) = |A| = a_{11} a_{22} a_{33}$

$$A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 4 & -2 & 0 & 0 \\ -5 & 6 & 1 & 0 \\ 1 & 5 & 3 & 3 \end{bmatrix}$$

main diagonal الدياجونال

is $|A| = 2 \times -2 \times 1 \times 3 = -12$

بعض اشكال ال determinate بتطلع صفر علطول ①

① تلاقي صف كله اصفار

$$0 = \begin{bmatrix} 0 & 0 & 0 \\ 2 & 4 & -5 \\ 3 & -5 & 2 \end{bmatrix}$$

② تلاقي عمود كله اصفار

③ تلاقي صفين زي بعض بالظبط او عمودين

$$0 = \begin{bmatrix} 1 & -2 & 4 \\ 0 & 1 & 2 \\ 4 & -2 & 4 \end{bmatrix}$$

④ العمود الاول مضروب العمود الثاني

$$\begin{array}{c} x-3 \\ \end{array}$$

$$0 = \begin{bmatrix} 1 & 2 & -3 \\ 2 & -1 & -6 \\ -2 & 0 & 6 \end{bmatrix}$$

If A is an $n \times n$ invertible matrix

$$\det(A^{-1}) = \frac{1}{\det(A)}$$

$$|A^{-1}| = \frac{1}{|A|}$$

If a matrix B results from a matrix A by interchanging two rows (columns) of A, then

$$|B| = -|A|$$

عند تبديل صفين او تبديل عمودين يتم الضرب في إشارة سالبة

e.g. $\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = - \begin{vmatrix} 3 & 4 \\ 1 & 2 \end{vmatrix}$

$$\begin{vmatrix} 2 & 1 & -3 \\ 4 & 0 & 1 \\ 0 & 0 & 2 \end{vmatrix} = - \begin{vmatrix} 1 & 2 & -3 \\ 0 & 4 & 1 \\ 0 & 0 & 2 \end{vmatrix}$$

$|AB| = |A||B| = |B||A|$

$|A+B| \neq |A| + |B|$

If $|AB^2| = I_n \Longrightarrow |A||B|^2 = |I_n| = 1$

$\longrightarrow |A| \neq 0$

is nonsingular

$$\begin{vmatrix} 1 & -3 & -4 & -2 \\ -1 & 4 & 3 & 4 \\ 2 & -5 & 1 & 8 \\ 0 & 1 & 0 & -1 \end{vmatrix}$$

$$1 \begin{vmatrix} 4 & 3 & 4 \\ -5 & 1 & 8 \\ 1 & 0 & -1 \end{vmatrix} \neq 5 \begin{vmatrix} 3 & 4 \\ 0 & -1 \end{vmatrix}$$

5

Notes

$|AB| = |A|\,|B|$

$|ABC\cdots| = |A|\,|B|\,|C|\ldots|\ldots|$

$|A+B| \neq |A| + |B|$

$|A^n| = |A|^n$

A square matrix A is invertible (non singular) if and only if $|A| \neq 0$

If ~~A~~ A and B are 5x5

$|A| = 8 \qquad |B| = 2$

$|A^2| = |A|^2 = 8^2 = 64$

$|-A| = (-1)^5 |A| = -8$

$|A^T B^{-1}| = |A^T|\,|B^{-1}| = |A| \cdot \frac{1}{|B|} = 4 = 8 \times \frac{1}{2}$

$|2A^{-1}B^4A| = 2^5 |A^{-1}|\,|B^4|\,|A| = 32\,\frac{1}{|A|}\,|B^4|\,|A|$

$32\,|B^4| = 512$

Part 13

Cramer's Rule

$$x_1 = \frac{\det(A_1)}{\det(A)}, \quad x_2 = \frac{\det(A_2)}{\det A}, \quad \ldots \quad x_n = \frac{\det(A_n)}{\det(A)}$$

$a_{11}x_1 + a_{12}x_2 = b_1$
$a_{21}x_1 + a_{22}x_2 = b_2$

$$|A| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \qquad |A_1| = \begin{vmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{vmatrix}$$

and $|A_2| = \begin{vmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{vmatrix}$

$$x_1 = \frac{|A_1|}{|A|}, \quad x_2 = \frac{|A_2|}{|A|}$$

$4x_1 - 2x_2 = 10$
$3x_1 - 5x_2 = 11$

$$|A| = \begin{vmatrix} 4 & -2 \\ 3 & -5 \end{vmatrix} = (4)(-5) - (3)(-2) = -20 + 6 = \boxed{-14}$$

$$|A_1| = \begin{vmatrix} 10 & -2 \\ 11 & -5 \end{vmatrix} = (10)(-5) - (11)(-2) = -50 + 22 = \boxed{-28}$$

$$A_{3\times3} = \begin{pmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{pmatrix}$$

$$M_{11} = \begin{vmatrix} A_{22} & A_{23} \\ A_{32} & A_{33} \end{vmatrix}$$

Compute the cofactor matrix C

$$= \begin{pmatrix} \overset{+}{C_{11}} & \overset{-}{C_{12}} & \overset{+}{C_{13}} \\ \overset{-}{C_{21}} & \overset{+}{C_{22}} & \overset{-}{C_{23}} \\ \overset{+}{C_{31}} & \overset{-}{C_{32}} & \overset{+}{C_{33}} \end{pmatrix}$$

, where $C_{ij} = (-1)^{i+j}(M_{ij})$

$C_{11} = (-1)^2 M_{11}, \ldots C_{12} = (-1)^3 M_{12}$

The adjoint matrix $(A) = C^T$

Part 19

$\vec{V}$

vector in the Plane

vector ٣ طرق لكتابة
Terminal Point
Initial Point (origin) نقطة الاصل

$$x = (x_1, x_2) = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

vectors in R^2 vectors in R^n

R^2 = 2-space = set of all ordered pair of real numbers

a 2-vector $u = (-3, 5) = \begin{pmatrix} -5 \\ 5 \end{pmatrix} = \begin{bmatrix} -3 \\ 5 \end{bmatrix}$

in vector R^2 or vector in the plane

vector in R^3 = 3-space = set of all ordered triples of real numbers

a 3-vector $v = (1, 0, -2)$

$\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$ is a vector in R^3

or vector in the space الفضاء

vectors in R^n

R^n = n-space = set of all ordered n-tubles of real numbers

An n-vector $v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$ is called a vector in R^n

where $v_1, v_2, \dots, v_n \in R$ are called the components of the vector

Part 14 ~~Part 20~~

adjoint Matrix

منقوله مفروقة الماصلات الكاحبا

steps: Given Matrix $A_{n\times n}$

① Find the minor M_{ij} for each element A_{ij}
② compute the Cofactor matrix C
③ The adjoint matrix $adj(A) = C^T$

/ /

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Minor a_{21}

$$M_{21} = \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}$$

Cofactor of a_{21}

$$C_{21} = (-1)^{2+1} M_{21} = -M_{21}$$

$$A = \begin{array}{c} \begin{array}{ccc} + & - & + \end{array} \\ \begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{bmatrix} \end{array}$$

$$|A| = 2 \begin{vmatrix} 3 & 2 \\ 4 & 1 \end{vmatrix} + 1 \begin{vmatrix} 3 & -1 \\ 4 & 0 \end{vmatrix}$$

$$-2 \times -5 + 4 = 14$$

$$\det(A) = |A| = 14$$

[Part 15]

Determinants

$$|A| = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = a_{11}\,a_{22} - a_{21}\,a_{12}$$

$$A = \begin{bmatrix} 2 & -3 \\ 1 & 2 \end{bmatrix}$$

$$\det(A) = |A| = 2 \times 2 - 1 \times -3 = \boxed{7}$$

The Determinants 3X3 matrix

$$A = \begin{bmatrix} \overset{+}{a_{11}} & \overset{-}{a_{12}} & \overset{+}{a_{13}} \\ \overset{-}{a_{21}} & \overset{+}{a_{22}} & \overset{-}{a_{23}} \\ \overset{+}{a_{31}} & \overset{-}{a_{32}} & \overset{+}{a_{33}} \end{bmatrix}$$

$$|A| = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

determinant of a 3x3 Matrix

~~a11 a12 a13 a11 a~~

$$\begin{array}{ccccc} a_{11} & a_{12} & a_{13} & a_{11} & a_{12} \\ a_{21} & a_{22} & a_{23} & a_{21} & a_{22} \\ a_{31} & a_{32} & a_{33} & a_{31} & a_{32} \end{array}$$

$$-a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{33}a_{21}a_{12}$$

$$a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32}$$

$+$ $|A|$

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{bmatrix}$$

$$\begin{array}{ccccc} & & -1\times & & \\ & & -4 & 0 & 6 \\ 0 & 2 & 1 & 0 & 2 \\ 3 & -1 & 2 & 3 & -1 \\ 4 & 0 & 1 & 4 & 0 \\ & & 0 & 16 & 0 \end{array}$$

$$|A| = 0 + 16 + 0 - (-4) - 0 - 6 = 14$$

محاضرة

Linear vector space

(x_1, y_1, z_1)
(x_2, y_2, z_2)

Linear vector space → فراغ اتجاهي خطي

$$\begin{array}{r} (1,2,3) \\ +\ (4,5,6) \\ \hline (5,7,9) \end{array}$$

subspace → فراغ اتجاهي خطي جزئي

span of set → المجموعة المنشرة للفراغ

① Linear vector space is set of V

① $u+v \in V$
② $u+v = v+u$ (Commutative)
③ $u+(v+w) = (u+v)+w$ (associative)
④ there is a unique element, denoted by 0, such that $u+0=u$ for every $u \in V$
⑤ there is a unique element, denoted by $-u + (-u) = 0$ where 0 is zero element of V

الانغلاق

⑥ $\alpha u \in V$

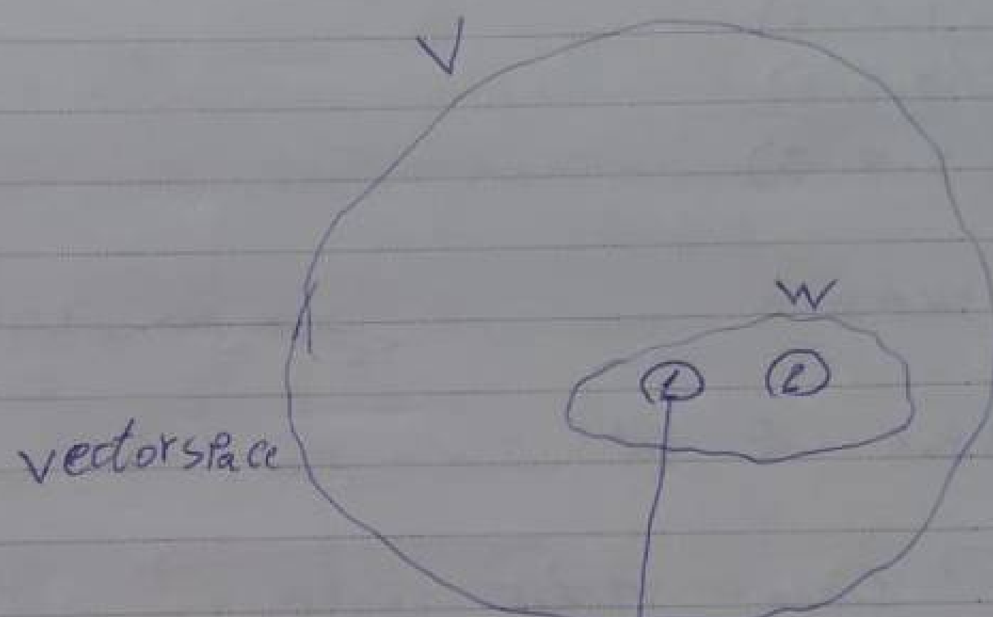
⑦ $\alpha(u+v) = \alpha u + \alpha v$

⑧ $\alpha(\beta u) = (\alpha\beta) u$

⑨ $(\alpha+\beta)u = \alpha u + \beta u$

⑩ $1 \cdot u = u$

sub space



خاصية الانغلاق الجمع والضرب X عدد ثابت

$u+v \in W \quad , \forall\, u, v \in W$

$c \cdot u \in W \quad ,$

$$\begin{pmatrix} 1 & 8 & 11 \\ 1 & 2 & 1 \\ -1 & 1 & 4 \end{pmatrix}$$

Span of set vector

$$(1,2,3) = (1,0,0) + 2(0,1,0) + 3(0,0,1)$$

$\boxed{\text{Linear combinantion}}$

$$w = c_1 v_1 + c_2 v_2 + \cdots + c_n v_2$$

in R^3 (x,y,z) linear combinantion

$$e_1 = (1,0,0),\ e_2 = (0,1,0) \text{ and } e_3 = (0,0,1)$$

$V = [1\ 8\ 11]^T$ is a linear combination of

$V_1 = [1\ \ 2\ \ 1]$ and $V_2 = [-1\ \ 1\ \ 4]$

$$V = c_1 v_1 + c_2 v_2$$

$$\begin{pmatrix} 1 \\ 8 \\ 11 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} -1 \\ 1 \\ 4 \end{pmatrix}$$

$$= \begin{pmatrix} c_1 \\ 2c_1 \\ c_1 \end{pmatrix} + \begin{pmatrix} -c_2 \\ c_2 \\ 4c_2 \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 8 \\ 11 \end{pmatrix} = \begin{pmatrix} c_1 - c_2 \\ 2c_1 + c_2 \\ c_1 + 4c_2 \end{pmatrix} \longrightarrow \begin{matrix} c_1 - c_2 = 1 \\ 2c_1 + c_2 = 8 \\ c_1 + 4c_2 = 11 \end{matrix}$$

$$3c = 9 \Rightarrow c_1 = 3$$
$$c_2 = 2$$

$V_1 = c_1 v_1 + c_2 v_2$

$3 + 4(2) = 11$

Mara أمنى الحجار

$$\boxed{\text{Basis} = \text{span} + \text{independent}}$$

$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ $v_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ in R^2

$\begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix} = 1 \neq 0$ independent

$v_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$ $v_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ $v_3 \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$

$\begin{vmatrix} 1 & 1 & 2 \\ 2 & -1 & 1 \\ 0 & 1 & 1 \end{vmatrix} = 0$

dependent

not span

R^2 (x, y)

R^3 (x, y, z)

$P_2(x)$ $a_0 + a_1x + a_2x^2$

$P_3(x)$ $a_0 + a_1x + a_2x^2 + a_3x^3$

$M_{2\times2}(R)$ $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$

Linear Idependence

basis

~~[illegible]~~ الخطي

Liner Dependence ~~[illegible]~~ الاستقلال الخطي

$0 = c_1v_1 + c_2v_2 + c_3v_3 \cdots + c_nv_n$

$c_1 = c_2 = c_3 = \cdots = c_n = 0 \rightarrow$ indendent

dependent $\exists c \neq 0$

eigen values and Eigen vectors

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 6 \end{bmatrix} = 3 \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

$$\boxed{AX = \lambda X}$$

$$AX - \lambda X = 0$$

$(A - \lambda I) X = 0$ → المعادله المميزه

$\lambda = -2$ Let $x = \begin{pmatrix} x \\ y \end{pmatrix}$

$$A \begin{pmatrix} -4 & 12 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$-4x + 12y = 0$$
$$-x + 3y = 0$$

$$-x + 3y = 0$$
$$-x = -3y$$
$$x = 3y$$

$\begin{pmatrix} 3 \\ 1 \end{pmatrix}$ ايش الخطوة

independent property

$$A = \begin{bmatrix} 1 & 1 \\ -1 & 3 \end{bmatrix}$$

$$\begin{vmatrix} 1-\lambda & 1 \\ -1 & 3-\lambda \end{vmatrix}$$

$$\cancel{(1-\lambda)(3-\lambda)+1=0}$$

$$\lambda^2 - 4\lambda + 4 = 0$$

Finding the characeristic polynomial of a matrix

matrix $2\times2 = \lambda^2 - \underbrace{tra(A)}_{(\lambda_1+\lambda_2)}\lambda + \underbrace{det(A)}_{(\lambda_1 \lambda_2)} = 0$

matrix $3\times3 = \lambda^3 - tra(A)\lambda^2 + C\lambda - det(A) = 0$

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$\lambda^2 - \underbrace{4}_{\text{القطر الرئيسي}}\lambda + \underbrace{3}_{\text{قيمة المحدد}} = 0$$

$$A^2 - 4A + 3I = 0 \longrightarrow$$ المصفوفة تحقق المعادلة المميزة بتاعتها

المعكوس A^{-1}

$$A^{-1}A^2 - 4A^{-1}A + 3A^{-1}I = 0$$

$$A - 4I + 3A^{-1} = 0$$

$$\boxed{A^{-1} = \frac{1}{-3}(A - 4I)}$$ المعادلة المميزة

If $\alpha \in \mathbb{R}$, then $\lambda + \alpha$ is an eigenvalue of $A + \alpha I_n$

vectors in R^n

R^n = n-space = set of all ordered n-tuples of real num

An n-vector $v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$ is called a vector in R^n

where $v_1, v_2, v_3, \dots v_n \in R$ are called the **components** of the vector

vector Equality

Two vectors $u = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$ and $v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$

in R^n are said to be equal if $u_i = v_i$, $1 \leq i \leq n$

Exp

Two vectors $u = \begin{bmatrix} 2 \\ 0 \\ 5 \end{bmatrix}$ and $v = \begin{bmatrix} 2 \\ 0 \\ 5 \end{bmatrix}$

in R^3 are equal ✓

Two vectors $s = (-3, 2, 0, 7)$ $w = (-3, 2, 0, 7)$
in R^4 are equal ✓

vector spaces

#19

vector in the plane

by directed line segment

origin نقطة بداية

(x_1, x_2) terminal نقطة نهاية

vectors are represented by lower case ($u, v, w,$ and x)

٣ طرق لكتابة ال vectors

$$x = (x_1, x_2) = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

vectors in R^n ← عدد صحيح ← الاعداد الحقيقية

٢١٩٨

لو اكبر من R^2 يسمى ب vector in the space

اقل من او يساوي R^2 ب vector in the plane

vectors in R^2

R^2 = 2-space = set of all orderd pairs of real num

a 2-vector $u = (-3, 5) = \begin{pmatrix} -3 \\ 5 \end{pmatrix} = \begin{bmatrix} -3 \\ 5 \end{bmatrix}$ is vector in R^2 or vector in the plane

R^3 = 3-space = set of all orderd triple of real num

a 3-vector $v = (1, 0, -2) = \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$ is a vector in R^3

vector in the space

ALADIB

Vector Addition

$$u = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \quad \text{and } v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \quad \text{in } R^n$$

$$u + v = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{bmatrix}$$

$$u = \begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix} \quad v = \begin{bmatrix} -2 \\ 2 \\ -3 \end{bmatrix}$$

$$u + v = \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix} \quad \text{in } R^3$$

Vectors المتجهات

Let u, v and w be vectors in $\mathbb{R}^n$, and let c and d be scalars

$u+v$ is a vector in $\mathbb{R}^n$

$u+v = v+u$

$(u+v)+w = u+(v+w)$

$u+0=u$

$u+(-u)=0$ is the zero vector in $\mathbb{R}^n \rightarrow \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$

عددهم n

cu is a vector in $\mathbb{R}^n$

$c(u+v) = cu+cv$

$(c+d)u = cu+du$

$c(du) = (cd)u$

$1(u) = u$

Mezen / /

vector spaces

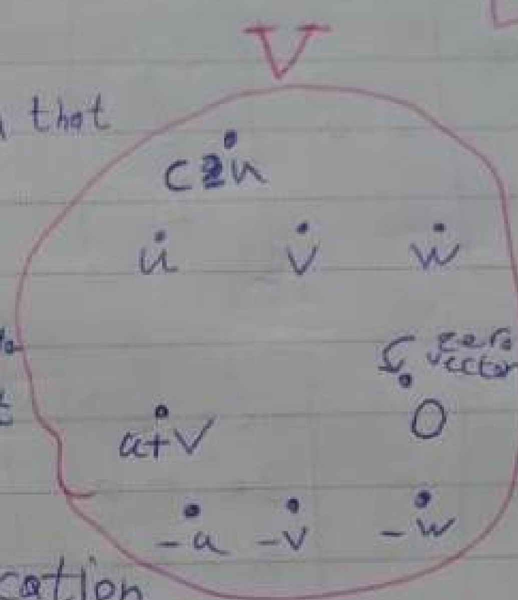
21#

فضاء المتجهات

vector addition and scalar multiplication

Addition

1. $u+v$ is in V
2. $u+v = v+u$
3. $u+(v+w) = (u+v)+w$
4. V has a zero vector 0 such that for every u in V, $u+0=u$
5. For every u in V, there is a vector in V denoted by $-u$ such that $u+(-u)=0$



لو طلع بره

يبقى ليس vector space

scalar multiplication

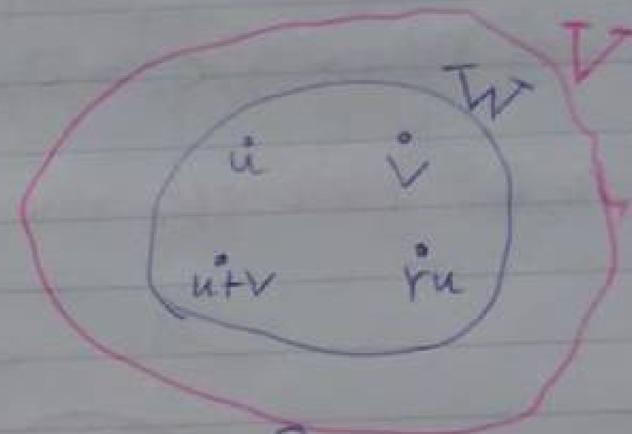
6. cu is in V
7. $c(u+v) = cu+cv$
8. $(c+d)u = cu+du$
9. $c(du) = (cd)u$
10. $1(u) = u$

Subspace #22

الفضاء الجزئي

If u and v are in W, then $u+v$ is in W

If u is in W and c is any scalar, then cu is in W



$r \in R$

Spanning sets

Let $S = \{v_1, v_2, \ldots v_k\}$ be a subset of a vector space V

$$\operatorname{span}(S) = \{c_1 v_1 + c_2 v_2 + \cdots + c_k v_k : c_1, c_2, \ldots, c_k \in R\}$$

Maram

#24

Linear Independence

Linearly independent

$$c_1 v_1 + c_2 v_2 + \cdots + c_k v_k = 0$$ zero vector

has only the trivial solution

$c_1 = 0, c_2 = 0, \ldots, c_k = 0$

If there are also nontrivial soln, then is

Linearly dependent

$S = \{v_1, v_2, v_3\} = \{(1,2,3), (0,1,2), (-2,0,1)\}$

$c_1 v_1 + c_2 v_2 + c_3 v_3 = 0$

$c_1(1,2,3) + c_2(0,1,2) + c_3(-2,0,1) = (0,0,0)$

$$\left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 2 & 1 & 0 & 0 \\ 3 & 2 & 1 & 0 \end{array}\right] \xrightarrow[R_3 = -3R_1 + R_3]{R_2 = -2R_1 + R_2} \left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 1 & -4 & 0 \\ 0 & 2 & 7 & 0 \end{array}\right]$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 2 \end{bmatrix}$$

$$\xrightarrow[R_3 = 2R_2 + R_3]{R_1 = -5R_2 + R_1} \left[\begin{array}{ccc|c} 1 & 0 & -\frac{2}{3} & 0 \\ 0 & 1 & \frac{1}{3} & 0 \\ 0 & 0 & \frac{5}{3} & 0 \end{array}\right] \xrightarrow{R_3 = \frac{3}{5} R_3}$$

$$\begin{bmatrix} 1 & 0 & -\frac{2}{3} \\ 0 & 1 & \frac{1}{3} \\ 0 & 0 & 1 \end{bmatrix} \longrightarrow$$

VIP

$Det(A) = 0$ dependent

not span

$Det(A) \neq 0$ independent

span

Basis and Dimension

#25

A set of vectors $S = \{v_1, v_2, \dots v_k\}$ be a subset of a vector space R^n is said to form a basis for a subspace W of R^n iff

① $W = \text{span}(S)$

② S is linerly independent set

let $W = \left\{ \begin{bmatrix} a-b \\ 2a \\ 3a+4b \end{bmatrix} a, b \in R \right\}$ be a subset of the vector space R^3

$$\begin{bmatrix} a-b \\ 2a \\ 3a+4b \end{bmatrix} = \begin{bmatrix} a \\ 2a \\ 3a \end{bmatrix} + \begin{bmatrix} -b \\ 0 \\ 4b \end{bmatrix} = a \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + b \begin{bmatrix} -1 \\ 0 \\ 4 \end{bmatrix}$$

$$W = \left\{ a \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + b \begin{bmatrix} -1 \\ 0 \\ 4 \end{bmatrix} ; a, b \in R \right\} = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 4 \end{bmatrix} \right\}$$

W is a subspace of R^3